From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA INTEGRATED TEST SERIES

PART TEST – IV

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 198**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above MUST be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 08)** contains 8 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (09 to 14) contains 3 paragraphs with each having 2 questions. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (15 to 20) contains 6 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

Single Correct Choice Type

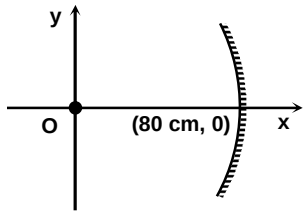
This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. A man in a boat is looking at the bottom of a lake of depth H. What will be he apparent depth of the lake if the line of vision of man forms an angle r with the vertical

(A) $(H/\mu)\sec r$ (B) H/μ

(C) $\frac{H}{\mu}\frac{\cos^2 r}{1-\left(\sin^2 r/\mu^2\right)^{3/2}}$ (D) None of these

2. In a three dimensional coordinate system (OXYZ), a concave mirror of radius of curvature 40 cm is placed at x = 80 cm. An object placed at origin is given a velocity $\vec{v}_0 = \left(9\hat{i}+6\hat{j}+3\hat{k}\right)$ cm/s. Find out the magnitude of velocity of its image.



(A) $\sqrt{46}$ cm/s (B) $\sqrt{6}$ cm/s

(C) $\frac{\sqrt{14}}{3}$ cm/s (D) None of these

3. It is desired to have a combination of two lenses placed very close to each other such that effective focal length of the combination for all the colours is same then

(A) Both must be biconvex lenses
(B) Both must be biconcave lenses
(C) At least one lens must be plano-concave lens
(D) If one lens is a converging lens then the other must be a diverging lens

4. For an equilateral prism, it is observed that when a ray strikes grazingly at one face it emerges grazingly at the other. Its refractive index will be

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{2}{\sqrt{3}}$

(C) 2 (D) Data not sufficient

Space for rough work

5. In a standard Young's double slit experiment performed with light of a single wavelength, it is observed that the difference between the intensity at the maximum and the minimum is (1/3) of the maximum intensity. If this is due to a difference in the amplitude of the light waves emerging from each slit, the ratio of the amplitudes is

(A) 3 : 2 (B) $\sqrt{3} : \sqrt{2}$

(C) 2 : 1 (D) none of these

6. If a physical quantity A associated with a hydrogen atom satisfies:

$\frac{A}{\sqrt{r}}$ = constant, where r is the radius of the electronic orbit of the hydrogen atom (not necessarily in the ground state), then A could be

(A) wavelength of emitted radiation from the state

(B) frequency emitted radiation from the state

(C) Angular momentum of a revolving electron in the state

(D) Energy of the emitted radiation from the state

7. In a series LCR circuit, the frequencies f_1 and f_2 at which the current amplitude falls to $1/\sqrt{2}$ of the current amplitude at resonance are separated by frequency interval

(A) R/L (B) $2\pi\sqrt{LC}$

(C) $\sqrt{LC}$ (D) RC

8. Two indentical glass plates are kept close and parallel to each other. Each glass plate reflectes 20 percent of light incident on it and rest is transmitted. The ratio of the minimum and maximum intensities in the interference pattern formed by the two beams obtained after one reflection at each plate is

(A) $\frac{1}{25}$ (B) $\frac{1}{49}$

(C) $\frac{1}{64}$ (D) $\frac{1}{81}$

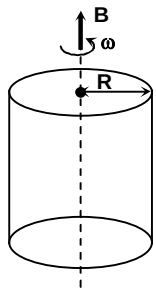
Space for rough work

Linked Comprehension Type

*This section contains 3 paragraphs. Based upon each paragraph, 2 multiple choice questions have to be answered. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.*

Paragraph for Questions 9 and 10

A very long solid conducting cylinder of radius R is placed in a constant and uniform magnetic field B which is parallel to the axis of the cylinder. Now the cylinder is rotated with a constant angular velocity ω as shown in the figure. Electronic charge and mass can be taken as e and m respectively.



9. The electric field inside the cylinder as a function of radial distance r is

(A) $E(r) = \left(\omega B - \frac{m\omega^2}{e}\right)r$ (B) $E(r) = \frac{m\omega^2}{e}r$

(C) $E(r) = \omega Br$ (D) $E(r) = \left(\omega B + \frac{m\omega^2}{e}\right)r$

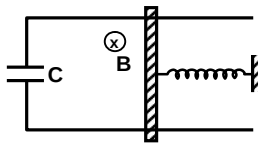
10. The potential difference between the axis of the cylinder and circumference is

(A) $\Delta V = \left(\omega B - \frac{m\omega^2}{e}\right)\frac{R^2}{2}$ (B) $\Delta V = \frac{m\omega^2R^2}{2}$

(C) $\Delta V = \frac{\omega BR^2}{2}$ (D) $\Delta V = \left(\omega B + \frac{m\omega^2}{e}\right)\frac{R^2}{2}$

Space for rough work

Paragraph for Question Nos. 11 to 12



A straight resistanceless conductor of mass m is placed on two parallel resistance less smooth rails as shown in the figure. Distance between both the rails is ℓ. Leftmost ends of both the rails is shorted with the help of a capacitor C. The conductor is joined to a non-conducting ideal spring of spring constant k which is fixed to a rigid support. The whole system is placed in a static magnetic field B which is directed into the plane of the paper.

Now the conductor is shifted to right by a distance x_0 to compress the spring and the conductor is released from the position.

11. Find the least time to reach the conductor to its original position, where the spring was undeformed.

(A) $\frac{\pi}{2}\sqrt{\frac{m - CB^2\ell^2}{k}}$ (B) $\frac{\pi}{2}\sqrt{\frac{m}{k}}$

(C) $\frac{\pi}{2}\sqrt{\frac{CB^2\ell^2}{k}}$ (D) $\frac{\pi}{2}\sqrt{\frac{m + CB^2\ell^2}{k}}$

12. Find the maximum speed of the conductor.

(A) $x_0\sqrt{\frac{k}{m + CB^2\ell^2}}$ (B) $x_0\sqrt{\frac{k}{m - CB^2\ell^2}}$

(C) $x_0\sqrt{\frac{k}{m}}$ (D) $x_0\sqrt{\frac{k}{CB^2\ell^2}}$

Space for rough work

Paragraph for Question Nos. 13 to 14

Uranium $^{238}_{92}U$ is an unstable nucleus. It decays to Thorium $^{238}_{92}Th$, which is again an unstable nuclei which further decays to $^{234}_{91}Pa$. Such decay is called successive disintegration. Let $^{238}_{92}U$ is called as A of decay constant λ_1 and $^{234}_{90}Th$ is called as B of decay constant λ_2 and stable nuclei $^{234}_{91}Pa$ is named as C. Here A is called parent nucleus and B is called daughter nucleus of A. Any two adjacent nuclei may be consider parent or daughter nuclei. Suppose N_1, N_2 and N_3 are number of nuclei A, B and C respectively at time t which undergoes decay. Then we can write

$$A \xrightarrow{\lambda_1} B \xrightarrow{\lambda_2} C$$

Rate of disintegration of A = $\frac{dN_1}{dt} = -\lambda_1 N_1$

Rate of disintegration of B = $-\lambda_2 N_2$ and net rate of formation of B = $\frac{dN_2}{dt} = \lambda_1 N_1 - \lambda_2 N_2$

Rate of formation of nuclei C is equal to $\frac{dN_3}{dt} = \lambda_2 N_2$

If at t = 0, there are N_0 number of nuclei of A, where as nuclei B and C are absent in the sample answer the following questions

13. Number of nuclei of B at any time t is

(A) $\frac{N_0\lambda_1}{\lambda_2-\lambda_1}\left(e^{\lambda_1 t} - e^{\lambda_2 t}\right)$ (B) $\frac{N_0\lambda_2}{\lambda_2-\lambda_1}\left(e^{-\lambda_1 t} - e^{-\lambda_2 t}\right)$

(C) $\frac{N_0\lambda_1}{\lambda_2-\lambda_1}\left(e^{-\lambda_1 t} - e^{-\lambda_2 t}\right)$ (D) $\frac{N_0\lambda_2}{\lambda_2-\lambda_1}\left(e^{-\lambda_1 t} + e^{-\lambda_2 t}\right)$

14. Number of nuclei of nuclei C at time t is

(A) $N_0\left[1+\frac{\lambda_1}{\lambda_2-\lambda_1}e^{-\lambda_2 t} - \frac{\lambda_2}{\lambda_2-\lambda_1}e^{-\lambda_1 t}\right]$ (B) $N_0\left[\frac{\lambda_1}{\lambda_2-\lambda_1}e^{-\lambda_2 t} - \frac{\lambda_2}{\lambda_2-\lambda_1}e^{-\lambda_1 t}\right]$

(C) $N_0\frac{\lambda_1}{\lambda_2-\lambda_1}\left(e^{-\lambda_2 t} - e^{-\lambda_1 t}\right)$ (D) $N_0\left[\frac{\lambda_1}{\lambda_2-\lambda_1}e^{-\lambda_2 t} - \frac{\lambda_2}{\lambda_2-\lambda_1}e^{-\lambda_1 t} - 1\right]$

Multiple Correct Answer(s) Type

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.

15. Choose correct statement regarding X-ray

(A) If the target material in a Coolidge tube is changed keeping the acceleration voltage constant, the minimum wavelength of the X-rays produced will change.
(B) X-ray do not get deflected on application of electric or magnetic fields.
(C) X-ray can ionize a gas.
(D) X-ray can be reflected, refracted, diffracted and polarized under suitable condition.

Space for rough work

16. Choose the correct statement(s)
(A) The density of nuclear matter is independent of the size of the nucleus.
(B) The binding energy per nucleon, for nuclei of middle mass numbers, is about 8 MeV.
(C) A free neutron is unstable.
(D) A free proton is stable.

17. A lens of focal length 'f' is placed in between an object and screen at a distance 'D'. The lens forms two real images of object on the screen for two of its different positions, a distance 'x' apart. The two real images have magnifications m_1 and m_2 respectively ($m_1 > m_2$). Then

(A) $f = \frac{x}{m_1 - m_2}$ (B) $m_1 m_2 = 1$

(C) $f = \frac{D^2 - x^2}{4D}$ (D) $D \geq 4f$.

18. A thin paper of thickness 0.02 mm having refractive index 1.45 is pasted across one of the slits in a Young's double slit experiment. The paper transmits 4/9 of light falling on it.
(λ_{light} = 600 nm).
(A) Amplitude of light wave transmitted through the paper will be 2/3 times that of incident wave.
(B) The ratio of maximum and minimum intensity in the fringe pattern will be 25.
(C) The total number of fringe crossing the centre if an identical paper is pasted on the other slit is 15.
(D) The ratio of maximum and minimum intensity in the pattern will be 5.

19. A neutron having kinetic energy E collides head on with a stationary hydrogen atom in ground state, which is initially at rest but free to move. Assume the mass of the neutron to be equal to that of H atom. Use the energy of the nth state for H atom as $E_1 = -13.6$eV; $E_2 = -3.4$eV; $E_3 = -1.5$eV. (whether collision is elastic or inelastic, depends on the energy of the neutron, and in case of elastic collision they can be treated as ordinary particles) now choose the correct options:
(A) The collision will be perfectly elastic if E is less than 20.4 eV.
(B) If E = 24.2 eV, the H atom may get excited to n = 4.
(C) If E < 20.4eV the kinetic energy of the neutron will become zero after the collision.
(D) If E > 20.4eV the kinetic energy of the neutron will become zero after the collision.

20. A ray of light travelling in air falls on the surface of water in such a way that the reflected ray is completely plane polarised. The refractive index of water for the given ray is 4/3. Choose the correct option/options.
(A) The maximum angular deviation of the ray is 74°
(B) The minimum angular deviation of the ray is 16°
(C) The maximum angular deviation of the ray is 106°
(D) The minimum angular deviation of the ray 106°

Space for rough work

Chemistry

PART – II

SECTION – A

Straight Objective Type

This section contains **8 multiple choice** questions numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which only one is correct.

1. A crystal is made of particles A, B and C. A forms FCC packing B occupies all octahedral voids of A and C occupies all tetrahedral void of A. If all the particles along one body diagonal are removed them the formula of the crystal would be
   (A) ABC_2 (B) A_2BC_2
   (C) $A_8B_4C_5$ (D) $A_5B_4C_8$

2. Blister copper is
   (A) ore of Cu (B) pure Cu
   (C) alloy of Cu (D) Cu containing 2% impurity

3. H_2SO_4 is added while preparing a standard solution of Mohr' salt to prevent
   (A) hydration (B) reduction
   (C) hydrolysis (D) complex formation

4. In the reaction, the products P and Q are
   $Au + HCl + HNO_3 \rightarrow P + Q + H_2O$
   (A) $HAuCl_4$, NO_2 (B) $AuCl_3$, NO_2
   (C) $AuCl_3$, N_2O_4 (D) $HAuCl_4$, N_2O_5

5. Which of the following can be separated by H_2S in dil HCl?
   (i) $Bi^{3+}Mn^{2+}$ (ii) $Al^{3+}Hg^{2+}$
   (iii) $Zn^{2+}Cu^{2+}$ (iv) $Ni^{2+}Cu^{2+}$
   (A) (i) and (ii) (B) (i) and (iii)
   (C) (i), (ii), (iii) (D) All

Rough work

6. The vapour pressure of a solution of 6.69 gm of $Ca(NO_3)_2$ in 100 gm water is 746.9 torr at 373 K. What is degree of dissociation of the salt?
(A) 0.68 (B) 0.32
(C) 0.18 (D) 0.46

7. Consider the given data:

Half – cell reaction	standard reduction potential E° (volts)
1. $Cr_2O_7^{-2} + 14H^+ + 6e^- \rightleftharpoons 2Cr^{3+} + 7H_2O$	1.33V
2. $Cr^{3+} + 3e^- \rightleftharpoons Cr$	-0.74V
3. $Cu^+ + e^- \rightleftharpoons Cu$	0.52 V
4. $Cu^{2+} + 2e^- \rightleftharpoons Cu$	0.34 V

The numerical value of the standard cell potential for the reaction
$2Cr_{(s)} + 3Cu^{2+}_{(aq)} \rightleftharpoons 2Cr^{3+}_{(aq)} + 3Cu_{(s)}$ is
(A) - 1.08 volts (B) - 0.40 volts
(B) 1.08 volts (D) 2.50 volts

8. The product of reaction of an aqueous solution of Bi^{3+} salt with sodium thiosulphate gives
(A) Bi_2S_3 (B) $Na_3[Bi(S_2O_3)_3]$
(C) $Na[Bi(S_2O_3)_2]$ (D) $[Bi_2(S_2O_3)_2]Cl_2$

Rough work

Linked Comprehension Type

*This section contains 3 paragraphs. Based upon each paragraph, 2 multiple choice questions have to be answered. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.*

Paragraph for question Nos 9 to 10

The electrical energy produced by a galvanic cell is given by the product of its electromotive force and the quantity of electricity which passes. Till about the middle of the nineteenth century it was generally believed that electrical energy of a reversible cell originated from the decrease in the enthalpy ($-\Delta H$) of the cell reaction. This view received support from the fact that the enthalpy change of the reaction

$$Zn(s) + Cu^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Cu(s)$$

taking place in the Daniell cell is about 210,000 joules which is very close to the electrical energy of the cell viz., 212,300 joules, as calculated above. However, towards the close of the century, it was pointed out by Gibbs, and independently by Helmholtz, that the electrical energy of a reversible cell originates from the free energy decrease ($-\Delta G$) of reaction occurring in the cell. Suppose in a particular cell reaction, n is the number of electrons liberated at one electrode (or taken up at the other electrode), then evidently, n faradays (nF coulombs) of electricity will be generated in the complete cell reaction. If, for the sake of simplicity, the EMF of the cell, viz., E_{cell} is denoted by E, then

Electrical energy produced by the cell = nFE

Hence, $-\Delta G = nFE$

Read the following paragraph and answer the questions given below:

9. In the Daniell cell, the EMF is 1.10 volts. The cell reaction involves liberation as well as taking up of 2 moles of electrons. What will be the electrical energy generated in the cell for the complete cell reaction

(A)106, 150 joule (B) 212, 300 joule

(C) 53, 075 joule (D) None of these

10. For the Daniel cell involving the cell reaction,

$Zn(s) + Cu^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Cu(s)$

The standard free energies of formation of Zn(s), Cu(s), $Cu^{2+}(aq)$ and $Zn^{2+}(aq)$ are 0, 0, 64.4 kJ mol^{-1} and –154.0 kJ mol^{-1}, respectively. Calculate the standard EMF of the cell.

(A) 1.13 volt (B) 2.10 volt

(C) 0.5 volt (D) 1.95 volt

Rough work

Paragraph for question Nos 11 to 12

Addition of non-volatile solute always lowers the vapour pressure of solvent, therefore, it will be in equilibrium with solid phase at a lower pressure and hence at a lower temperature, the difference between the freezing points of the pure solvent and its solution is called depression of freezing point. The depression in freezing point when 1 mol of solute is dissolved in 1000 gm of solvent is called molal freezing point depression constant (K_f). The value of K_f for H_2O is 1.86 K mol^{-1}kg. To understand this, a solution of 25% (w/w) $MgCl_2$ containing impurity of the chloride of a trivalent metal M having atomic mass 43.5 gm (24% by weight of $MgCl_2$) has been prepared. In this solution $MgCl_2$ is completely dissociated. On the other hand only 50% dissociation of MCl_3 takes place and molality is approximately equal to molarity.

Read the following paragraph and answer the questions given below:

11. Depression in freezing point of the solution is
    (A) 17.36 K (B) 13.02 K
    (C) 14.88 K (D) 12.42 K

12. If the solution is cooled to –21.7°C, what weight of ice will be formed?
    (A) 15 gm (B) 60 gm
    (C) 40 gm (D) 51 gm

Paragraph for Question Nos. 13 to 14

One pungent colourless gas “A” and another colourless gas “B” both are soluble in water, produce a solid substance “C”. At high temperature when gas “A” is passed through heated sodium a solid substance “D” and H_2 gas is produced. When aqueous solution of “B” is added to a solution of $AgNO_3$ a white precipitate of “E” is obtained.

Read the above paragraph carefully and answer the questions given below it:

13. Pungent colourless gas A is:
    (A) HCl (B) NH_3
    (C) PH_3 (D) H_2S

14. After the reaction of A and B, compound C is formed; compound C is
    (A) NH_2NH_2 (B) NH_4Cl
    (C) $H_2S + Cl_2$ (D) PH_4Cl

Rough work

Multiple Correct Answers Type

This section contains **6 multiple correct** answer(s) type questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE** is/are correct.

15. Cataphoresis is the process used for the coagulation of colloids. It can be used for
(A) colloids having positive charge
(B) colloids having negative charge
(C) lyophilic colloids
(D) All of the above

16. The solid compound SiO_2 exist in the forms of
(A) crystalline in quartz
(B) amorphous in quartz
(C) amorphous in silica glass
(D) crystalline in silica glass

17. Given $E^o_{Ag^+|Ag} = 0.80V, E^o_{Mg^{2+}|Mg} = -2.37V, E^o_{Cu^{2+}|Cu} = 0.34V, E^o_{Hg^{2+}|Hg} = 0.79V$. Which of the following statement(s) is/are incorrect?
(A) $AgNO_3$ can be stored in copper vessel.
(B) $Cu(NO_3)_2$ can be stored in magnesium vessel.
(C) $CuCl_2$ can be stored in silver vessel.
(D) $HgCl_2$ can be stored in copper vessel.

18. Identify the correct statement(s)
(A) 1 M NaCl has higher freezing point than 1 M glucose solution.
(B) 1 M NaCl solution has almost same boiling point as 1 M KCl solution, assuming 100% ionisation.
(C) Apparent molecular weight of NaCl will be less than 58.5 in water because it undergoes dissociation.
(D) i = 1 when acetic acid is dissolved in benzene.

19. Which of the following is/are true?
(A) All halogens exist as diatomic gaseous molecules at ordinary temperature
(B) bromine dissolves in chloroform imparting blue colour
(C) chlorine monoxide is evolved when $KClO_3$ is treated with conc. H_2SO_4
(D) chloride ion is confirmed by chromyl chloride test

20. Which of the following is/are correct for lyophillic aquasols?
(A) Its surface tension is lower than that of H_2O.
(B) Its viscosity is higher than that of water.
(C) Colloidal silver
(D) Colloidal antimony

Rough work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains **8 multiple choice questions** numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. Two squares are chosen from the squares of an ordinary chess board. It is given that the selected square do not belong to the same row or column. The probability that the chosen squares are of same column is equal to

(A) $\frac{25}{49}$ (B) $\frac{32}{49}$

(C) $\frac{25}{64}$ (D) $\frac{1}{2}$

2. Area bounded by the curve y = sinx, y = cosx, x = $-\frac{\pi}{3}$, x = 2π

(A) $4\sqrt{2}-\left(\frac{\sqrt{3}+1}{2}\right)$ (B) $\sqrt{2}+\left(\frac{\sqrt{3}+1}{2}\right)$

(C) $\sqrt{2}-\left(\frac{\sqrt{3}+2}{2}\right)$ (D) $4\sqrt{2}+\left(\frac{\sqrt{3}+1}{2}\right)$

3. If f(x) = x – 1 and g(x) = |f(|x|) – 2|, then the area bounded by y = g(x) and the curve $x^2 - 4y + 8 = 0$ is equal to

(A) $\frac{4}{3}\left(4\sqrt{2}-5\right)$ (B) $\frac{4}{3}\left(4\sqrt{2}-3\right)$

(C) $\frac{8}{3}\left(4\sqrt{2}-3\right)$ (D) $\frac{8}{3}\left(4\sqrt{2}-5\right)$

Rough work

4. Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ and $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ be three non-zero vectors such that $\vec{c}$ is a unit vector perpendicular to vectors $\vec{a}$ and $\vec{b}$. If the angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{6}$, then $\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}^2$ is equal to

(A) 0 (B) 1

(C) $\frac{1}{4}|\vec{a}|^2|\vec{b}|^2$ (D) $\frac{3}{4}|\vec{a}|^2|\vec{b}|^2|\vec{c}|^2$

5. Shortest distance between the lines $\vec{r} = (\hat{i} + 2\hat{j} - 2\hat{k}) + \lambda(\hat{j} + \hat{k})$ and $\vec{r} = (\hat{i} - \hat{j} - \hat{k}) + \mu(\hat{j} + \hat{k})$; $\lambda, \mu \in R^+$ is

(A) $\frac{1}{2\sqrt{3}}$ (B) $\sqrt{\frac{2}{3}}$

(C) $\sqrt{2}$ (D) $\frac{4}{\sqrt{3}}$

6. The area bounded by the curve $y = \frac{3}{|x|}$ and $y + |2 - x| = 2$ is

(A) $\frac{4 - \ln 27}{2}$ (B) $2 - \ln 3$

(C) $2 + \ln 3$ (D) none of these

7. The probability that the length of a randomly chosen chord of a circle lies between $\frac{2}{3}$ and $\frac{5}{6}$ of its diameter is

(A) $\frac{5}{16}$ (B) $\frac{1}{16}$

(C) $\frac{1}{4}$ (D) $\frac{5}{12}$

Rough work

8. If p, q, r are in A.P., then the determinants $\begin{vmatrix} a^2+2^{n+1}+2p & b^2+2^{n+2}+3q & c^2+p \\ 2^n+p & 2^{n+1}+q & 2q \\ a^2+2^n+p & b^2+2^{n+1}+2q & c^2-r \end{vmatrix}$ is equal to

(A) 1 (B) 0
(C) $a^2b^2c^2 - 2^n$ (D) $(a^2 + b^2 + c^2) - 2^nq$

Comprehension Type

This section contains **3 paragraphs.** Based upon the paragraph 2 multiple choice questions have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE IS CORRECT.**

Paragraph for Question Nos. 09 to 10

Read the following write up carefully and answer the following questions:

Two lines whose equations are $\frac{x-3}{2} = \frac{y-2}{3} = \frac{z-1}{\lambda}$ and $\frac{x-2}{3} = \frac{y-3}{2} = \frac{z-2}{3}$ lie in the same plane, then

9. Point of intersection of the lines lies on
(A) 3x + y + z = 20 (B) 3x + y + z = 25
(C) 3x + 2y + z = 24 (D) none of these

10. Equation of plane containing both lines is
(A) x + 5y – 3z = 10 (B) x + 6y + 5z = 20
(C) x + 6y – 5z = 10 (D) none of these

Paragraph for Question Nos. 11 to 12

Read the following write up carefully and answer the following questions:

A 3×3 determinant has its entries 1 or –1. The number of such determinant is $2^9 = 512$. We will call a 3×3 determinant with entries 1 or –1 as minus special if product of elements of any rows or any column is –1 for example $\begin{vmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{vmatrix}$ is a minus special determinant.

11. The number of $n \times n$ minus special determinant must be
(A) 2^{n-1} (B) $2^{(n-1)^2}$
(C) 2^{n+1} (D) $\frac{13n^2 - 37n + 26}{n}$

12. The minimum value of 3×3 minus special determinant is
(A) –2 (B) –4
(C) –6 (D) 0

Rough work

Paragraph for Question Nos. 13 to 14

Read the following write up carefully and answer the following questions:
If $A_1, A_2,\ldots..,A_n$ are mutually exclusive and exhaustive events and A is an event which together with one of $A_1, A_2\ldots..,A_n$, then $P(A) = P(A_1).P(A/A_1) + P(A_2) \cdot P(A/A_2) + \ldots.+P(A_n) \cdot P(A/A_n)$.

13. A ball is drawn from a box containing 8 white and 12 green balls and kept aside. The probability that another ball drawn is green is

(A) $\frac{4}{5}$ (B) $\frac{2}{3}$

(C) $\frac{1}{5}$ (D) $\frac{3}{5}$

14. A lot contains 20 articles. The probabilities that the lot contains exactly 2 and 3 defective articles are 0.4 and 0.6 respectively. Articles are drawn from the lot at random one by one without replacement and tried till all the defective are found. Then the probability that the testing producer ends at the 12th testing is

(A) $\frac{1}{19}$ (B) $\frac{98}{1900}$

(C) $\frac{101}{1900}$ (D) $\frac{99}{1900}$

Multiple Correct Choice Type

This section contains **6 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct.

15. If A and B are two independent events such that $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{5}$, then

(A) $P(A \cup B) = \frac{3}{5}$ (B) $P(A/B) = \frac{1}{2}$

(C) $P(A/A \cup B) = \frac{5}{6}$ (D) $P\left(\frac{A \cap B}{\overline{A} \cap \overline{B}}\right) = 0$

16. If A and B are two events such that $P(A) = \frac{1}{2}$ and $P(B) = \frac{2}{3}$, then

(A) $P(A \cup B) \geq \frac{2}{3}$ (B) $P(A \cap B') \leq \frac{1}{3}$

(C) $\frac{1}{6} \leq P(A \cap B) \leq \frac{1}{2}$ (D) $\frac{1}{6} \leq P(A' \cap B') \leq \frac{1}{2}$

Rough work

17. Let $\vec{a}, \vec{b}, \vec{c}$ are 3 vectors such that $\vec{a}\cdot\vec{a} = \vec{b}\cdot\vec{b} = \vec{c}\cdot\vec{c}$ and $|\vec{a}-\vec{b}|^2 + |\vec{b}-\vec{c}|^2 + |\vec{c}-\vec{a}|^2 = 27$ then

(A) $\vec{a}, \vec{b}, \vec{c}$ are necessarily coplanar

(B) $\vec{a}, \vec{b}, \vec{c}$ represents sides of a triangle in magnitude of direction

(C) $\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a}$ has the least value

(D) $\vec{a}, \vec{b}, \vec{c}$ represents mutually orthogonal vectors

18. Let E and F are independent event then

(A) E, F are mutually exclusive

(B) E, $\bar{F}$ are independent

(C) $\bar{E}$, $\bar{F}$ are independent

(D) $P\left(\frac{E}{F}\right) + P\left(\frac{\bar{E}}{F}\right) = 1$

19. Let PN be the perpendicular from the point P(1, 2, 3) to xy–plane. If OP makes an angle α with positive direction of the z–axis and ON makes an angle β with the +ve direction of x–axis, where O is the origin (α and β are acute angle), then

(A) $\sin\alpha\sin\beta = \frac{2}{\sqrt{14}}$

(B) $\cos\alpha\cos\beta = \frac{1}{\sqrt{14}}$

(C) $\tan\alpha = \frac{\sqrt{5}}{3}$

(D) $\tan\beta = 2$

20. Let $\vec{a}$ and $\vec{b}$ are two given perpendicular vectors, which are non–zero. A vector $\vec{r}$ satisfying the equation $\vec{r}\times\vec{b} = \vec{a}$, can be

(A) $\vec{b} - \frac{\vec{a}\times\vec{b}}{|\vec{b}|^2}$

(B) $2\vec{b} - \frac{\vec{a}\times\vec{b}}{|\vec{b}|^2}$

(C) $|\vec{a}|\vec{b} - \frac{\vec{a}\times\vec{b}}{|\vec{b}|^2}$

(D) $|\vec{b}|\vec{b} - \frac{\vec{a}\times\vec{b}}{|\vec{b}|^2}$

Rough work